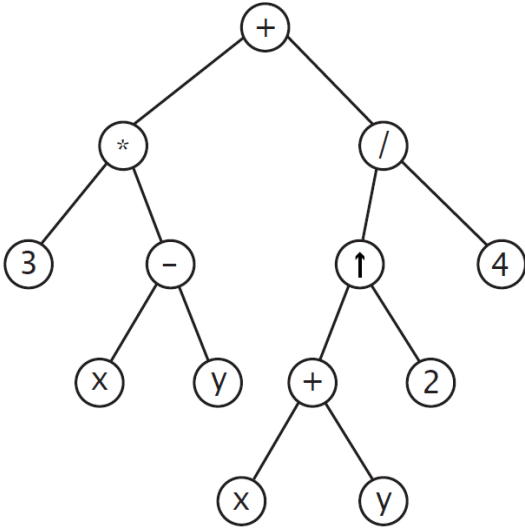
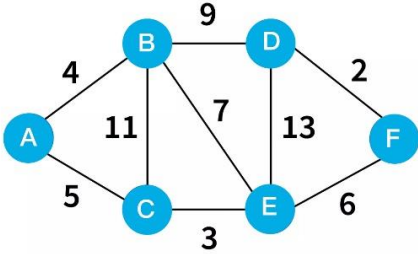
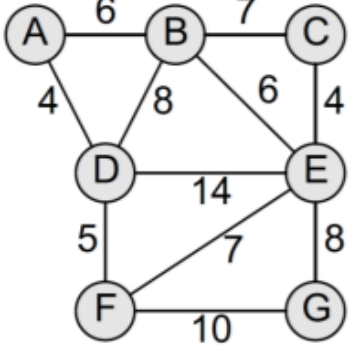
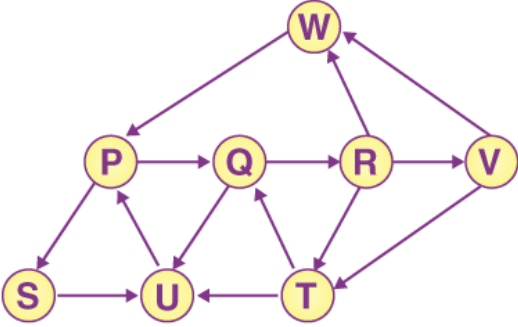


1.	a) Convert the following infix expression to equivalent postfix expression using stack data structure: $K + L - M \times N + (O \wedge P) \times W / V$ b) Consider the following queue of characters, where QUEUE is a circular array which is allocated six memory cells: FRONT=2, REAR=4 QUEUE: -, A, C, D, -, - Describe the queue as the following operations take place: i) F is added to the queue. ii) Two letters are deleted. iii) S is added to the queue. iv) One letter is deleted v) K, L, and M are added. vi) Two letters are deleted.	[6] [4]
2.	a) Mention the similarities between Merge sort and Quick sort algorithms. b) Derive the computation of worst case running time for Insertion-sort algorithm. c) Perform Binary Search for the array named $arr = \{-15, -22.5, -9, -4, -4.01, -2, -7.87\}$ and $key = -3.79$. (N.B.: You can do the necessary arrangement of the given array if required before performing Binary search algorithm.)	[3] [3] [4]
3.	a) The following represents file system hierarchy in Linux OS. We would like to add <i>jar.exe</i> file to /bin/ directory situated at the highest. Will this action change the height of the above hierarchy system? Also which order we should traverse the hierarchy in order to quickly access quickly <i>jar.exe</i> file? b) Find the value of the postfix expression: 6 2 ↑ 9 / 7 2 3 ↑ + 5 / +	[2 + 2] [3]

	c) Find the arithmetic expression (in postfix form) from the order rooted tree given below: 	[3]
4.	a) For the following graph, carry out Dijkstra's algorithm, starting from vertex B, to solve the single source shortest path problem.  b) Compute minimum spanning tree of the following graph using Prim's algorithms. 	[5] [5]
5.	a) Consider the following graph G.  Apply DFS graph traversal (starting from Q) on the given graph and label each vertex by the timestamp of discovering and finishing time.	[5] [5]

	b) Consider the following graph G. <pre>graph LR; a((a)) --> b((b)); b --> c((c)); c --> d((d)); d --> e((e)); f((f)) --> g((g)); g --> h((h)); h --> i((i)); i --> j((j)); a --> f; b --> g; c --> h; d --> i; e --> j; d --> i; i --> d; e --> j; j --> e;</pre> Apply BFS algorithm on it and compute shortest path distances of each vertex from d.																																					
6.	a) What do you mean by stable sorting algorithm? b) Apply counting-sort algorithm and show step by step procedure for the array {8, 5, 3, 8, 5} c) Mention the major difference about counting-sort compare with other sorting algorithms like Merge-sort, Quick-sort or Insertion-sort algorithms.	[2] [6] [2]																																				
7.	a) Briefly mention the procedure of how you can restore the heap property if a node violates the max heap property. (b) Build Max-Heap from the following numbers: {9, 15, 2, 18, 6, 13}	[3] [7]																																				
8.	a) Compute the Longest Common Subsequence (LCS) of the strings X = "AGGTAB" and Y = "GXTXAYB" using dynamic programming. Show how you calculated the LCS using a matrix containing both the cost and the arrows for each cell in the matrix. b) Suppose you have to allocate a room for activities whose start and end times are as follows. Compute the largest set of activities that can be arranged in this room using a greedy algorithm. Indicate the 1st, 2nd, 3rd, ... activities you choose using this algorithm and also indicate which activities you delete after you make each choice. <table><tr><td><i>i</i></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td><i>S_j</i></td><td>1</td><td>2</td><td>4</td><td>5</td><td>1</td><td>5</td><td>4</td><td>3</td><td>8</td><td>7</td><td>12</td></tr><tr><td><i>f_j</i></td><td>5</td><td>7</td><td>A</td><td>8</td><td>B</td><td>9</td><td>8</td><td>10</td><td>11</td><td>15</td><td>16</td></tr></table> where A = addition of the last two digits of your student ID B = addition of the last three digits of your student ID	<i>i</i>	1	2	3	4	5	6	7	8	9	10	11	<i>S_j</i>	1	2	4	5	1	5	4	3	8	7	12	<i>f_j</i>	5	7	A	8	B	9	8	10	11	15	16	[6] [4]
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N.B. Students must submit question along with answer scripts at the end of examination.

	c) Find the arithmetic expression (in prefix form) from the order rooted tree given below:	[3]
4.	a) For the following graph, carry out Dijkstra's algorithm, starting from vertex E, to solve the single source shortest path problem. b) Compute minimum spanning tree of the following graph using Prim's algorithm.	[5] [5]
5.	a) Consider the following graph G. Apply DFS graph traversal (starting from S) on the given graph and label each vertex by the timestamp of discovering and finishing time.	[5] [5]

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